November 13, 2026

To: Dr. Ljiljana Trajkovic, Editor-in-Chief
IEEE Transactions on Human-Machine Systems

Subject: New submission: "From Alerts to Actions: A Human–AI Decision System for Reliable Action Execution Under Disaster Risk"
Previous manuscript number: THMS-26-02-0115 (decision of 12 November 2026: Reject and Resubmit)

Dear Dr. Trajkovic,

We sincerely thank you, the Associate Editor, and the three reviewers for the careful evaluation of manuscript THMS-26-02-0115. As invited in your decision letter of 12 November 2026, we have considered every comment and now submit a substantially revised manuscript as a new submission.

The Associate Editor identified three main concerns: the novelty of the contribution, methodological transparency, and the alignment of the claims with the evidence. We agree with these concerns. In particular, we accepted Reviewer 3's observation about the Joint Cognitive Systems (JCS) literature and have fundamentally repositioned the paper. The revised manuscript no longer presents action-centered evaluation as a new concept. Instead, it presents a concrete application of established human factors principles to AI-based disaster decision support, together with an empirical evaluation.

The main changes are as follows.

1. **Positioning and contributions.** ACHDF is now explicitly framed as an instantiation, for disaster response, of the JCS perspective (Hollnagel & Woods, 2005; Woods & Hollnagel, 2006) and of adaptive function allocation (Parasuraman et al., 2000; Kaber & Endsley, 2004; Scerbo, 2007). A new Related Work subsection (II-A) explains how ACHDF relates to this body of work, and all three contributions have been rewritten.

2. **Definition of constructs.** A new Table I gives the role and operational definition of uncertainty, criticality, guardrails, bounded autonomy, escalation, trust, workload, and action execution, distinguishing design inputs, system modes, design considerations, and outcomes. Throughout the manuscript, we now state consistently that ACHDF adapts only the *level and form of AI support*, and that decision authority always remains with the human operator.

3. **Methodological transparency.** We now describe what each condition presented, the decision task and procedure, the participants, and all instruments (NASA-TLX, a three-item trust measure with consistent terminology, and the SUS). We explain that the Action Execution Rate (AER) combines action correctness and timeliness, and how it relates to response latency. Some of the requested details (scenario composition, interface screenshots, AER decomposition) could not be recovered from the available study materials. Rather than omit them silently, we list them explicitly in a new limitation (Section VII-D).

4. **Statistics.** A new Statistical Analysis subsection states which tests were used, what is reported, and what could not be computed. All outcomes are now reported together in a single table (Table V), with their means, differences, significance levels, and effect sizes (d_av) where computable. A post hoc sensitivity analysis addresses statistical

power. Because the trial-level data were not available for re-analysis, exact test statistics and confidence intervals could not be added; this is stated openly in the table note and in Section VII-D.

5. **Consistency and redundancy.** Numerical inconsistencies between the text and the tables have been corrected. Overlapping Results subsections (former V-A/V-B) were merged, redundant tables were consolidated into Tables V and VI, and the single Cohen's d whose comparison was unclear has been removed.

6. **Qualified claims.** Claims regarding mechanism, trust calibration, automation bias, the benefit of the action format, and cross-domain generalizability have been removed or reframed as hypotheses. Correspondingly, the Limitations section now addresses four points: the bundled nature of the intervention, the absence of a direct test of calibration and automation bias, domain and population generalizability, and the ecological validity of deferral.

7. **Structure and language.** The Results section now reports only descriptive and inferential findings. The former "mechanism analysis" has been moved to the Discussion and recast as a framework-based interpretation (Section VI-B). The manuscript has been edited for clarity and flow, inappropriate transitions ("in fact", "on the other hand") have been removed, and eleven references from the human factors literature have been added.

A detailed point-by-point response to the reviewers and a marked-up comparison of the original and revised manuscripts (latexdiff) are enclosed. We believe the revision has made the contribution and its evidential basis considerably clearer, and we would be grateful for your further consideration.

Sincerely,

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Enclosures:

1. Revised manuscript
2. Point-by-point response to reviewers
3. Marked-up comparison of original and revised manuscripts (latexdiff)